

Mechatronics Tutorial Sheet 2: Fourier Transform & Laplace Transform

1. Suppose that the discrete-time LTI systems shown in Figure 1 have frequency responses:

$$G(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega} + \frac{1}{4}e^{-2j\omega}}, \quad H(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$$

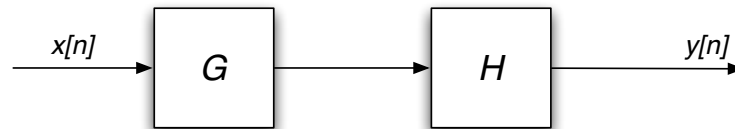


Figure 1

- a) Find a difference equation relating $x[n]$ to $y[n]$.
- b) Find the impulse response of the combined system.

a) $y[n] = x[n] * G[n] * H[n]$

$$\begin{aligned} \therefore \frac{Y(e^{j\omega})}{X(e^{j\omega})} &= G(e^{j\omega})H(e^{j\omega}) = \frac{2 - e^{-j\omega}}{(1 - \frac{1}{2}e^{-j\omega} + \frac{1}{4}e^{-2j\omega})(1 + \frac{1}{2}e^{-j\omega} - \frac{1}{4}e^{-2j\omega} + \frac{1}{8}e^{-3j\omega})} \\ &= \frac{2 - e^{-j\omega}}{1 + \frac{1}{8}e^{-3j\omega}} \end{aligned}$$

$$\therefore (1 + \frac{1}{8}e^{-3j\omega}) Y(e^{j\omega}) = (2 - e^{-j\omega}) X(e^{j\omega})$$

$$\therefore y[n] + \frac{1}{8}y[n-3] = 2x[n] - x[n-1]$$

b) for $x[n] = \delta[n] = \begin{cases} 1 & n=0 \\ 0 & \text{otherwise} \end{cases}$

assume that causality, $y[n] = 0$ for $n < 0$, therefore,

$$y[0] + 0 = 2 - 0 = 2 \quad \therefore y[0] = 2$$

$$y[1] + 0 = 0 - 1 = -1 \quad \therefore y[1] = -1$$

$$y[2] + 0 = 0 \quad \therefore y[2] = 0$$

$$y[3] + \frac{1}{8}y[0] = 0 \quad \therefore y[3] = -\frac{1}{8}y[0] = -\frac{1}{4}$$

$$y[4] + \frac{1}{8}y[1] = 0 \quad \therefore y[4] = -\frac{1}{8}y[1] = \frac{1}{8}$$

.....

Therefore, for $k = 0, 1, 2, 3, \dots$

$$\begin{cases} y[3k] = (-\frac{1}{8})^k y[0] = 2(-\frac{1}{8})^k \\ y[3k+1] = (-\frac{1}{8})^k y[1] = -(-\frac{1}{8})^k \\ y[3k+2] = 0 \end{cases}$$

2. For each of the following systems, plot the poles and zeros and determine whether or not the system is stable.

a) $H(s) = \frac{1}{s}$

b) $H(s) = \frac{1-s}{s^2-9}$

c) $H(s) = \frac{s}{s+3}$

d) $H(s) = \frac{3}{(s+1)(s+2)}$

e) $H(s) = \frac{s-2}{s+2}$

f) $H(s) = \frac{s+2}{s-2}$

g) $H(s) = \frac{1-s}{s^2+s+1}$

h) $H(s) = \frac{s^2+s+1}{(s+4)(s^2-s+1)}$

	zeros (s=)	poles (s=)	stable
a)	-	0	-
b)	1	± 3	x
c)	0	-3	✓
d)	-	-1 or -2	✓
e)	2	-2	✓
f)	-2	2	x
g)	1	$-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$	✓
h)	$-\frac{1}{2} \pm \sqrt{3}i$	-4 or $\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$	x

3. Suppose that $H(s)$ is the transfer function of a linear system with input $x(t)$ and output $y(t)$. Find $y(t)$ if:

- a) $H(s)$ is as in (2c), and $x(t)$ is an impulse.
- b) $H(s)$ is as in (2d), and $x(t)$ is a unit step.
- c) $H(s)$ is as in (2g), and $x(t)$ is a unit step.

a) $H(s) = \frac{s}{s+3}$, $x(t) = \delta(t)$

$$\therefore X(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = e^{-s0} = 1$$

$$\therefore Y(s) = H(s) \cdot X(s) = \frac{s}{s+3}$$

let $y(t) = \frac{d}{dt} z(t)$ $\therefore Y(s) = sZ(s)$ for $z(0) = 0$

$$\therefore Z(s) = \frac{1}{s+3}$$

$$\therefore z(t) = e^{-3t}$$

$$\therefore y(t) = \frac{d}{dt} z(t) = -3e^{-3t} \text{ for } t \geq 0 \text{ (causality)}$$

b) for $t \geq 0$,

$$H(s) = \frac{3}{(s+1)(s+2)} , x(t) = u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

$$\therefore X(s) = \int_0^{\infty} u(t) e^{-st} dt = \int_0^{\infty} e^{-st} dt = -\frac{1}{s} [e^{-st}]_0^{\infty} = \frac{1}{s}$$

$$\therefore Y(s) = X(s) \cdot H(s) = \frac{3}{s(s+1)(s+2)}$$

$$= \frac{A(s^2+3s+2)}{s(s+1)(s+2)} + \frac{B(s^2+2s)}{s(s+1)(s+2)} + \frac{C(s^2+s)}{s(s+2)(s+1)}$$

$$= \frac{3}{2} \frac{1}{s} - 3 \frac{1}{s+1} + \frac{3}{2} \frac{1}{s+2}$$

$$\therefore y(t) = \frac{3}{2} - 3e^{-t} + \frac{3}{2}e^{-2t}$$

c) for $t \neq 0$,

$$H(s) = \frac{1-s}{s^2+s+1} \quad \text{and} \quad X(s) = \frac{1}{s}$$

$$\begin{aligned}\therefore Y(s) &= H(s) \cdot X(s) = \frac{1-s}{s^2+s+1} \\&= \frac{A(s^2+s+1)}{s(s^2+s+1)} + \frac{Bs}{s(s^2+s+1)} \\&= \frac{1}{s} - \frac{s+2}{s^2+s+1} \\&= \frac{1}{s} - \frac{s}{s^2+s+1} - \frac{2}{s^2+s+1}\end{aligned}$$

Taking the inverse Laplace transform

$$\begin{aligned}y(t) &= 1 - s \frac{1}{2\pi j} \int_{6-j\infty}^{6+j\infty} \left(\frac{1}{s^2+s+1} \right) e^{-st} ds - 2 \frac{1}{2\pi j} \int_{6-j\infty}^{6+j\infty} \left(\frac{1}{s^2+s+1} \right) e^{-st} ds \\&= 1 - \frac{d}{dt} \left(\frac{e^{-\frac{1}{2}t}}{\sqrt{1-(\frac{1}{2})^2}} \sin(\sqrt{1-(\frac{1}{2})^2} t) \right) - 2 \left(\frac{e^{-\frac{1}{2}t}}{\sqrt{1-(\frac{1}{2})^2}} \sin(\sqrt{1-(\frac{1}{2})^2} t) \right) \\&= 1 - \frac{d}{dt} \left[\frac{2}{3} \sqrt{3} e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2} t\right) \right] - \frac{4}{3} \sqrt{3} e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2} t\right) \\&= 1 - \sqrt{3} e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2} t\right) + e^{-\frac{1}{2}t} \cos\left(\frac{\sqrt{3}}{2} t\right)\end{aligned}$$

4. a) Suppose that the signal $x(t)$ has Laplace transform $X(s)$. Show that

$$x(t - T) \xrightarrow{\mathcal{L}} e^{-Ts} X(s)$$

for $y(t) = x(t - T), \quad t \geq 0 \quad \therefore t - T \geq 0, \quad t \geq T$

$$Y(s) = \int_0^{\infty} y(t) e^{-st} dt = \int_0^{\infty} x(t - T) e^{-st} dt$$

Let $\tau = t - T \quad \therefore t = \tau + T$

$$\therefore Y(s) = \int_{0-T}^{\infty-T} x(\tau) e^{-s(\tau+T)} d\tau$$

$$= e^{-sT} \int_T^{\infty} x(\tau) e^{-s\tau} d\tau$$

$$= e^{-sT} X(s)$$

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5. a) Find the transfer function $\frac{Y(s)}{X(s)}$ for the system shown in Figure 5.1.

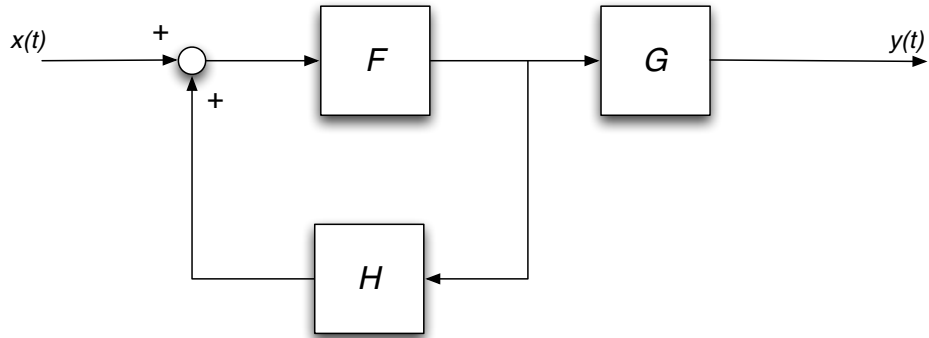


Figure 5.1

$$\frac{Y(s)}{X(s)} = \frac{F(s)G(s)}{1 - F(s)H(s)}$$

b) Find the transfer function $\frac{Y(s)}{X(s)}$ for the system shown in Figure 5.2.

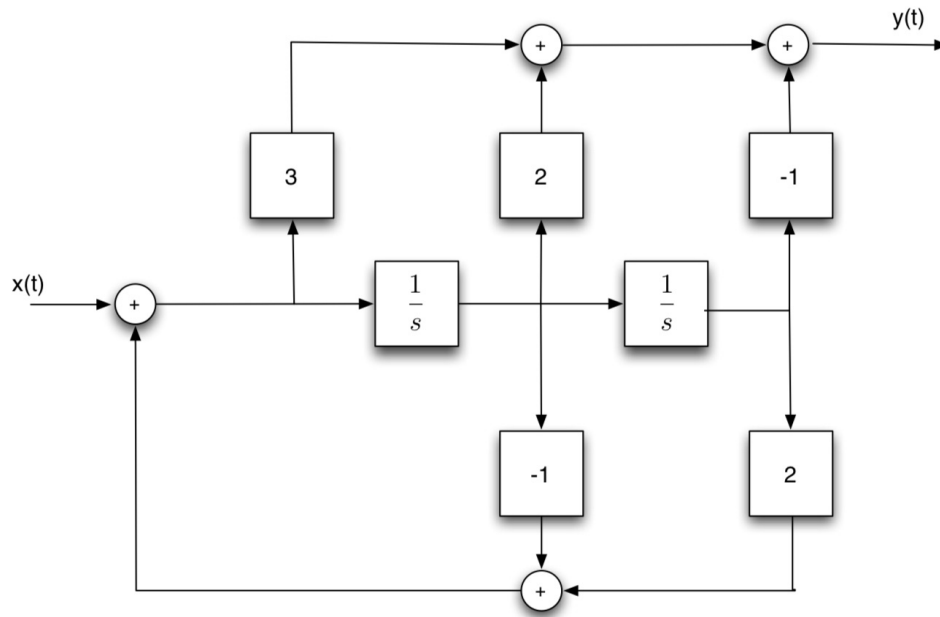


Figure 5.2

$$\begin{aligned} \frac{Y(s)}{X(s)} &= \frac{3 + (\frac{1}{s} \cdot 2) + [\frac{1}{s} \cdot \frac{1}{s} \cdot (-1)]}{1 - [\frac{1}{s} \cdot (-1) + \frac{1}{s} \cdot \frac{1}{s} \cdot 2]} \\ &= \frac{-\frac{1}{s^2} + \frac{2}{s} + 3}{-\frac{2}{s^2} + \frac{1}{s} + 1} \\ &= \frac{-1 + 2s + 3s^2}{-2 + s + s^2} \end{aligned}$$

c) In the system shown in Figure 5.3, find

- i) The transfer function from input to tracking error $\frac{e(s)}{x(s)}$
- ii) The transfer function from measurement noise to output $\frac{y(s)}{n(s)}$
- iii) The transfer function from disturbance to output $\frac{y(s)}{d(s)}$

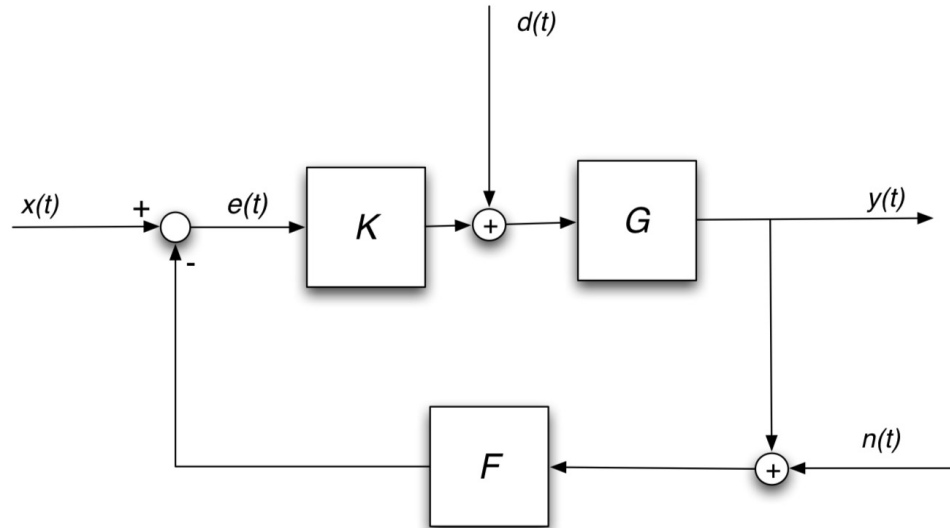


Figure 5.3

$$i) \frac{e(s)}{x(s)} = \frac{\cancel{-K(s)G(s)F(s)} + 1}{1 + K(s)G(s)F(s)}$$

$$ii) \frac{y(s)}{n(s)} = \frac{-K(s)G(s)F(s)}{1 + K(s)G(s)F(s)}$$

$$iii) \frac{y(s)}{d(s)} = \frac{G(s)}{1 + K(s)G(s)F(s)}$$

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6. Suppose that in Figure 5.3 the following transfer functions are given:

$$G(s) = \frac{s+1}{(s-2)(s+3)}, \quad F(s) = 1, \quad K(s) = K,$$

where K is a real number. Find the range of values of K for which the system is stable. Consider the input to be $x(t)$ and the output to be $y(t)$.

$$\begin{aligned} \frac{y(s)}{x(s)} &= \frac{K(s)G(s)}{1 + K(s)G(s)F(s)} \\ &= \frac{K(s+1)}{(s-2)(s+3) + K(s+1)} \\ &= \frac{K(s+1)}{s^2 + (1+K)s + (K-6)} \end{aligned}$$

for pole, $s^2 + (1+K)s + (K-6)$

$$\therefore s = \frac{-(1+K) \pm \sqrt{(1+K)^2 - 4(K-6)}}{2}$$

for $\text{Re}(s) < 0$.

$$K > 6$$